Final Year Project

CS6P05NI Project Proposal

5% Individual Coursework

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

Problem Scenario: Retail and student traders should have safe and low-cost learning environments concerning the learning of trading mechanics, strategy testing, and developing confidence without exposing capital. Paper-trading platforms that are currently in existence tend to lack both personalized learning and deliberate practice coupled with actionable feedback; they either replicate the user interface of a live broker with no pedagogy or provide automated backtests without step-by-step learning. The statistics of surveys and usage of the platform indicate that some beginners drop out due to their initial losses or lack of understanding; there is also a gap in education between studying courses and their application. The proposed project aims to fill that gap by integrating an interactive paper-trading experience, adaptive learning modules, strategysimulation and analytics to expedite the acquisition of skills.

The solution of the project: The given system is a web application that emulates the real market conditions with historical and simulated live data, offers a sandbox to place orders and has learning features: interactive tutorials, performance diagnostics and adaptive quizzes, a strategy lab where user can code, backtest and paper-run strategies. The application will provide learning paths based on the level (beginner to intermediate to advanced), tips provided in context when making a trade, and post-trade analytics that will help identify common errors and propose micro-lessons. This is a combination of an experimental trading sandbox and instructional scaffold to generate measurable learning outcomes.

2. Aims and Objectives

2.1 Aim:

With adaptive learning capabilities built in to allow the novice traders to advance to intermediate levels through practice, feedback and analytics, design a secure, responsive paper-trading web application.

2.2 Objectives (measurable):

- Deliver central paper-trading features (market feed simulation, limit/market orders, portfolio tracking) in 12 weeks.

- Prepare and incorporate a minimum of 12 interactive lessons and 8 adaptive quizzes on the topic of types of order, risk management, and strategy evaluation.

- Develop a strategy lab where you can upload/write simple strategy scripts, backtest with historical data and run it with simulated slippage on paper.

- Have a performance analytics engine which calculates metrics (win rate, expectancy, simulation returns Sharpe-like ratios) and produces automated learning recommendations.

- Perform a small user study (n 10 and above) and assess the improvement in trading knowledge and simulated performance of an account pre/post.

3. Expected Outcomes and Deliverables

- A fully functional web application (frontend + backend) deployed on a cloud test instance.

- Technical documentation: architecture diagrams, API specs, database schema, and deployment notes.
- User documentation: quick-start guide, tutorial script content, and assessment rubrics.

- Source code repository with README, tests, and CI setup.

- A short report summarizing user study results and recommended next steps.

4. Project Risk and Solutions

4.1 Risks:

- Availability of data: access to either live or historical market data can be rate-limited or restricted. Contingency: rely on freely available historical data (e.g. Yahoo Finance CSVs) and cache local data; in order to simulate realistic price series with adjustable volatility, as required.

Time: time overrun: The complexity of integrating learning/adaptive algorithms can be greater than schedule. Contingency: apply an MVP-first model to provide basic trading simulation and a bare minimum of lessons, and continue to improve.

- Security/privacy: data leakage or ineffective authentication by user. Contingency: use approved authorized libraries (OAuth2 or JWT), save only minimal user metadata and observe good security coding practices.

- Performance/scalability: high load resulting in slow performance on simulations. Contingency: maximize critical paths, also throttle heavy backtests on free tier, or use a job worker to queue tasks.

5. Methodology (Technical Approach & SDLC)

5.1 Approach:

Agile iterative development with two-week sprints. Key phases: requirements & design → implementation → testing → user evaluation → deployment.

5.1.1. High-level stack and SDLC tasks:

- Requirements & design: refine user stories and learning outcomes, create wireframes, and define APIs.

- Implementation: develop frontend (React), backend API (Node.js/Express or Python FastAPI), database (Postgres), and a worker queue (Redis + Celery or Bull) for backtests. Build a modular learning engine that maps performance metrics to lesson triggers.

- Testing: unit tests for logic, integration tests for APIs, end-to-end tests for core flows, and user acceptance testing with the study cohort.

- Deployment: containerized services (Docker), cloud hosting (Heroku/GCP/AWS light instance), CI/CD pipeline for builds and tests.

6. Resource Requirement

6.1 Hardware/software: development machine, cloud account for deployment (small VM), Docker, Postgres, Redis.

6.2 Libraries/services: React, Next.js (optional), FastAPI or Express, SQLAlchemy/TypeORM, plotting libraries for analytics (client-side), job queue (Celery/Bull), authentication provider (Auth0 or similar).

Datasets/publications: historical price feeds (Yahoo Finance, Alpha Vantage free tier), research on trading pedagogy and reinforcement learning for finance. Supervision: access to an academic supervisor for ethics approval (user study) and domain feedback.

7. WBS CODE

WBS CODE	Task/ Work package	Description	Duration	Deliverables
1.0	Project Initiation	Define project scope, stakeholders, constraints, risks, and feasibility	1 week	Project charter, scope document
1.1	Requirement Gathering	Identify all functional and non-functional requirements for the system	1 week	Requirement specification document
1.2	System Design Approval	Confirm and approve UI layout, system architecture, and project plan	1 week	Signed-off design and architecture blueprint
2.0	System Design	Create technical architecture, UI design, and database structures	2 weeks	System design documentation
2.1	UI Wireframe Design	Design mid-fidelity wireframes for dashboard, trading module, and learning module	1 week	Figma wireframes
2.2	Backend Architecture Design	Define API endpoints, data models, authentication flows, and database schema	1 week	Backend architecture and database schema

3.0	Backend Development	Implement core server logic and APIs	4 weeks	Backend services and endpoints
3.1	Authentication Module	Implement registration, login, JWT sessions, and user management	1 week	Auth API and user database
3.2	Paper Trading Engine	Implement order placement, trade simulation logic, and portfolio updating	2 weeks	Trading simulation API and core trading logic
3.3	Market Data Module	Integrate or simulate market price feeds for assets	1 week	Market data feed API
4.0	Frontend Development	Create user interface for all modules	4 weeks	Functional user interface
4.1	Dashboard UI	Display portfolio, market prices, charts, and alerts	2 weeks	Trading dashboard interface
4.2	Order Placement UI	Build the UI for buy/sell orders, order types, and validations	1 week	Order placement screen
4.3	Learning Module UI	Implement UI for tutorials, quizzes, and progress tracking	1 week	Learning content interface

5.0	Learning Engine Development	Implement adaptive learning rules and logic	3 weeks	Learning backend engine

5.1	Lesson Content Creation	Create educational content, quizzes, examples, and explanations	2 weeks	Lesson and quiz bank
5.2	Analytics Integration	Track user performance and suggest targeted lessons using analytics	1 week	Learning analytics engine
6.0	Backtesting & Strategy Tools	Develop tools to test trading strategies with historical data	3 weeks	Strategy lab and backtesting module
6.1	Backtest Engine	Code engine for historical simulation and performance results	2 weeks	Backtesting results generator
6.2	Strategy Editor	Provide UI for defining or selecting preset strategies	1 week	Strategy editor interface
7.0	Integration & Testing	Integrate modules and validate overall system functionality	3 weeks	Fully integrated system
7.1	Unit Testing	Test individual components and endpoints	1 week	Unit test reports
7.2	User Acceptance Testing (UAT)	Collect user feedback, measure usability, and refine system	2 weeks	UAT report and refinements

8.0	Deployment	Host and publish the system online	1 week	Live deployed application
8.1	Cloud Deployment	Configure database, hosting	1 week	Public URL and working deployment
		services, and secure access		
9.0	Documentation & Presentation	Produce technical and academic documentation	2 weeks	Complete project report and presentation
9.1	Technical Documentation	Write technical manual, API documentation, setup procedures	1 week	Technical documentation pack
9.2	Final Presentation	Prepare and deliver final presentation	1 week	Presentation slides and demo

8. Milestone

Milestone Code	Milestone	Completion Criteria	Expected Date
M1	Project Initiation Complete	Scope, problem definition, feasibility documented	Week 1
M2	Requirements Finalised	Functional & nonfunctional requirements approved	Week 2
M3	System Design Approved	Architecture, DB schema, and UI wireframes signed-off	Week 4
M4	Backend Core Completed	Authentication, trading engine & market API functional	Week 8
M5	Frontend Core Completed	Dashboard, order interface & learning UI working	Week 12
M6	Learning Engine Implemented	Lessons, quizzes & adaptive suggestions active	Week 15
M7	Backtesting Tools Ready	Strategy editor & backtest results operational	Week 18

9. Gantt Chart

Figure 1 Gantt chart

10. Conclusion

This coursework has successfully demonstrated the design and the project plan of a comprehensive paper trading web app filled with adaptive learning capabilities. With the integration of trading functionalities with interactive lessons, quizzes and performance based feedback, the system has addressed a vulnerable gap between theoretical financial education and practical real life trading experience. The scheduled and structured development plan of 24 weeks has included a systematic progression as it integrates requirement analysis and system design to implementation, testing, deployment and documentation.

The proposed solution not only allows users to practice trading in a risk free environment but also enriches their understanding of financial concepts and decision-making strategies through many available guided learning modules. Moreover, the addition of backtesting engine and analytics further empowers users to evaluate and refine their strategies that promotes evidence- based trading habits.

Therefore, the project is a huge contribution that serves as an educational tool for novice traders, students and financial learners. Its design makes it a strong foundation for future enhancements such as AI- driven recommendations, and other mobile platform expansion. Hence, the paper trading web app stands as a strong system that is capable of bridging the learning-practice divide in financial literacy and digital trading ecosystems.